Simple Types

Concepts of Programming Languages

CAS CS 320

Outline

- Have a high-level discussion of **type theory** in general
- Introduce and analyze the **simply-typed lambda calculus** (STLC)
- Demo an **implementation** of the STLC

Type Theory

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let f : int -> int = ...
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Types help us delineate "well-behaved" programs

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- Simplicity/Usability
- Expressivity
- Safety/Theoretical Guarantees

OCaml

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# let big_omega =
    let little_omega x = x x in
    little_omega little_omega;;
Error: This expression has type 'a -> 'b
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More expressive means more complex, designing PLs is finding the balance

Simply-Typed Lambda Calculus

Typing Judgments

Syntax (STLC)

Typing (STLC)

Example

$$\{x : \bot \to \bot\ , y : \bot\ \} \vdash \lambda z\,.\,z(xy) : (\ \bot \to \bot\ ) \to \bot$$

Example

$$\emptyset \vdash \lambda x \,.\, xx :$$

Type Annotations

Aside: Church vs. Curry Typing

```
fun x -> x

fun (x : unit) -> x
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(using Curry-style typing is not the same as having polymorphism)

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Every expression has a *unique type* in STLC with Church-style typing

In particular, the function `type_of` is well-defined

Recall: Semantics (STLC)

$$(1)\ \frac{}{\lambda x\,.\,e \Downarrow \lambda x\,.\,e}$$

$$(2)\ \frac{e_1 \Downarrow \lambda x\,.\,e \qquad e_2 \Downarrow v_2 \qquad [v_2/x]e \Downarrow v}{e_1 e_2 \Downarrow v}$$

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It doesn't determine **how** we evaluate the program

Practice Problem

$$\frac{(x:\tau) \in \Gamma}{\Gamma \vdash x : \tau} \qquad \frac{\Gamma, x:\tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}$$

$\cdot \vdash \lambda f^{\bot \to \bot}.\, \lambda x^{\bot}.\, fx : (\bot \to \bot) \to \bot \to \bot$

Give a derivation for the above judgment

Answer

$$\cdot \vdash \lambda f^{\top \to \top} . \lambda x^{\top} . fx : (\top \to \top) \to \top \to \top$$

How do we know if we've defined a "good" programming language?

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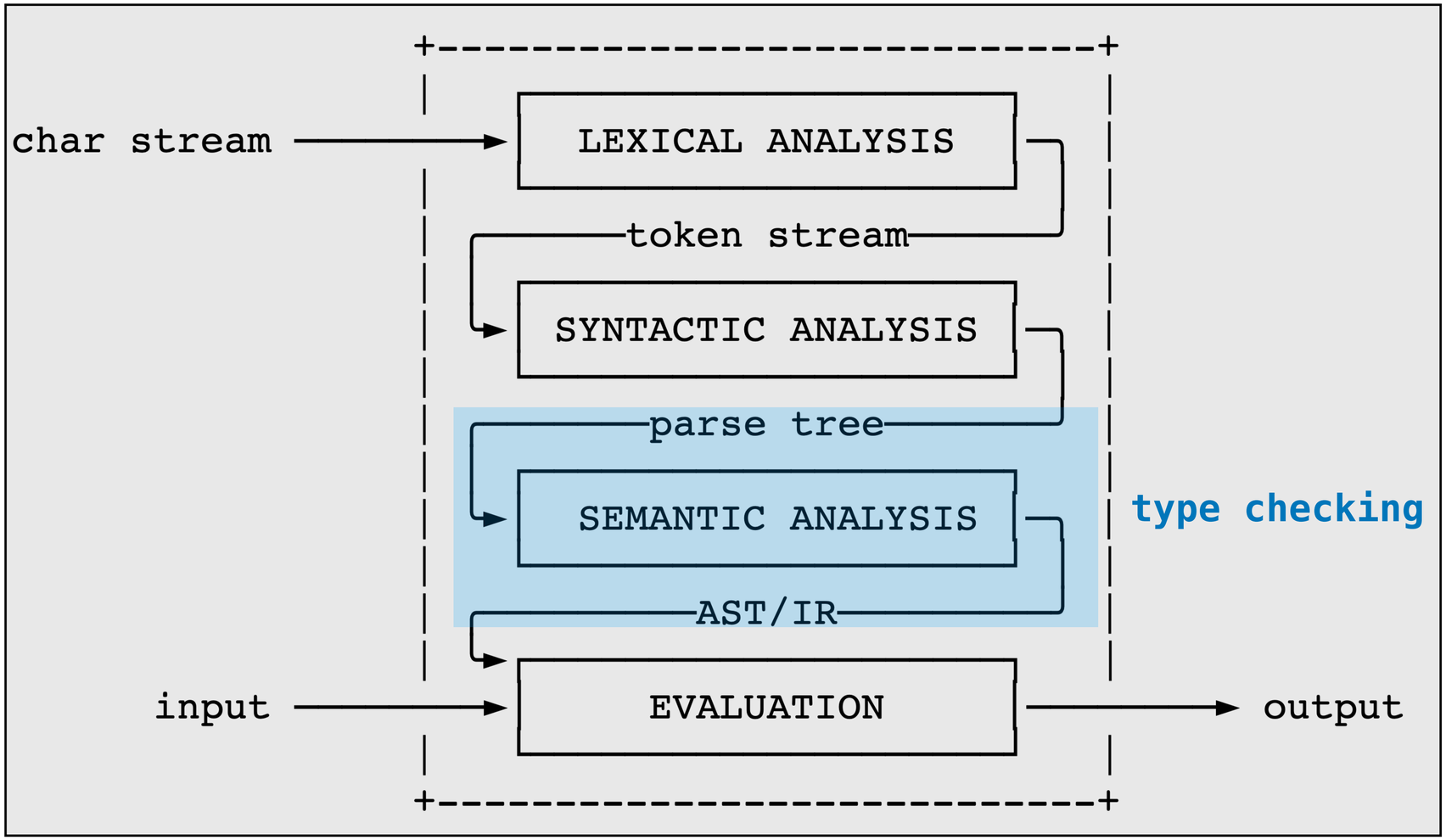
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These results are *fundamental*. They tell us that our PL is well-behaved (it's a "good" PL)

Type Checking

The Picture


char stream
LEXICAL ANALYSIS
token stream
SYNTACTIC ANALYSIS
parse tree
SEMANTIC ANALYSIS
type checking
AST/IR
input
EVALUATION
output

Type Checking vs. Type Inference

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For STLC, they are both easy

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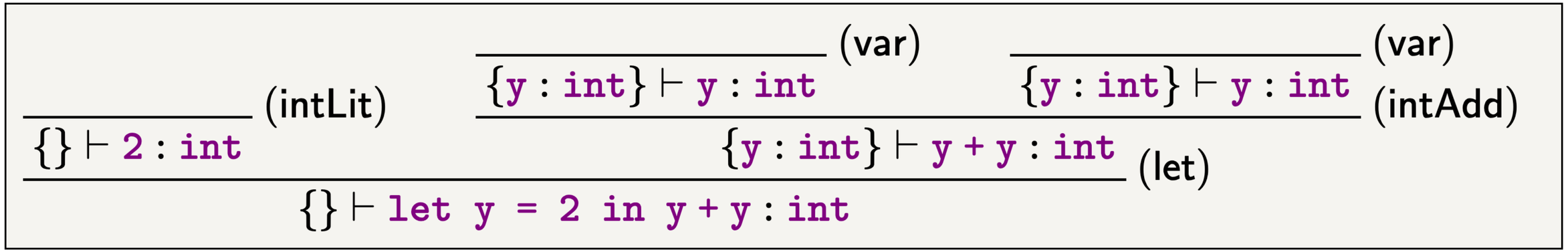
Our solution: We'll just use type inference

demo
(STLC)

Induction over Derivations

The Key Idea

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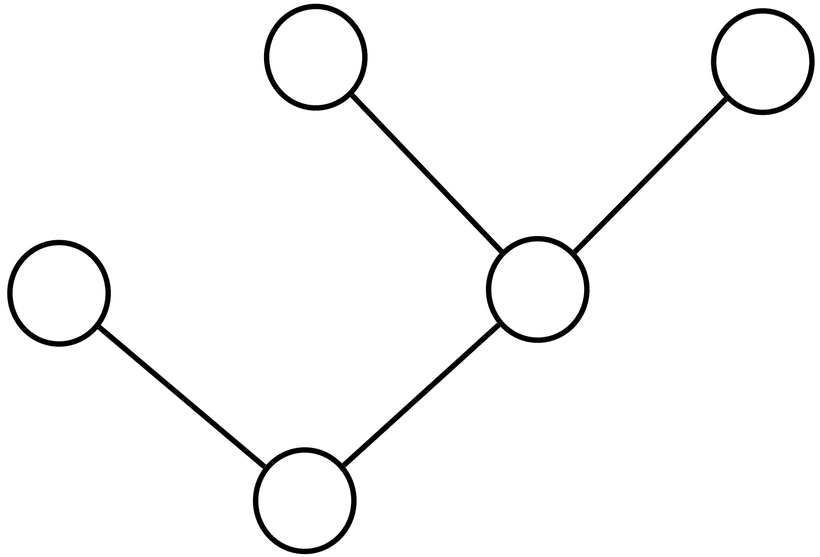
Derivations are *trees*

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We can prove things about trees using induction



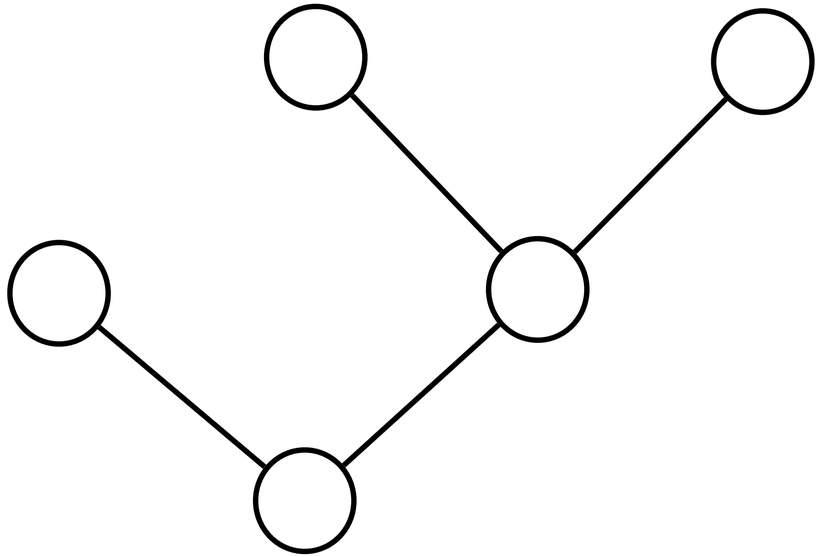
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We can prove things about trees using induction

We can prove things about *derivable judgments* using induction



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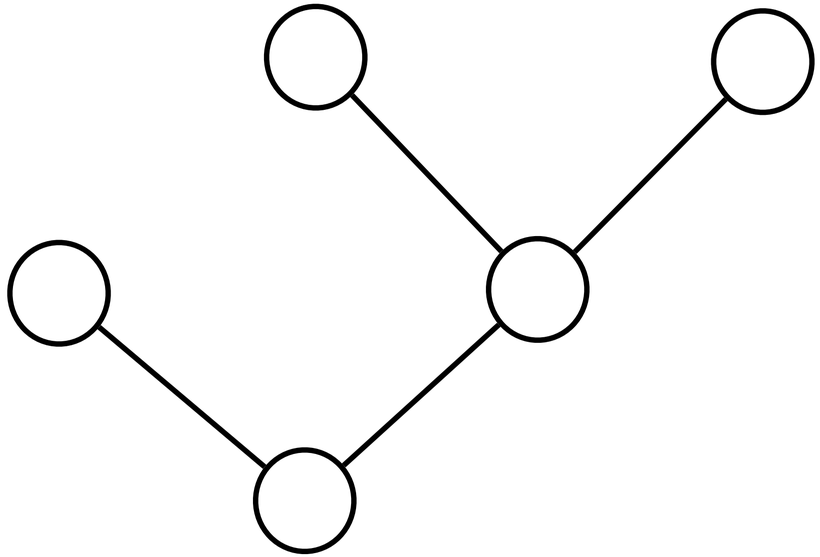
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We can prove things about *derivable judgments* using induction

Important: Every derivable judgment corresponds to a derivation



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> **Theorem.** If e is well-typed in Γ, then e is well-scoped

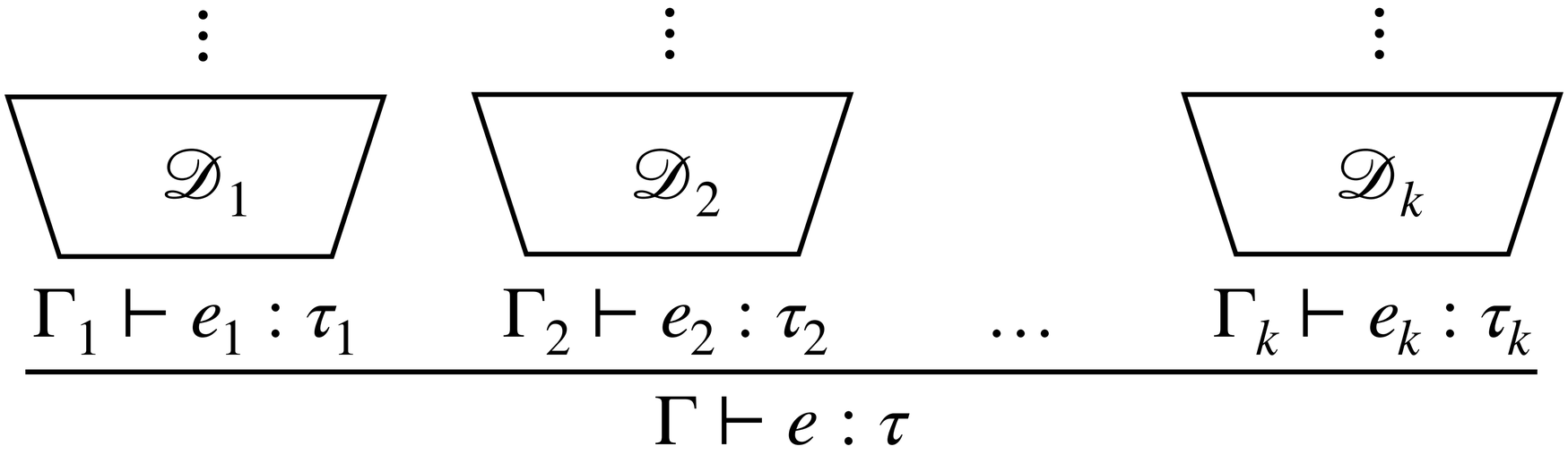
Proof. By induction on *derivations*

Base Case: Axioms

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

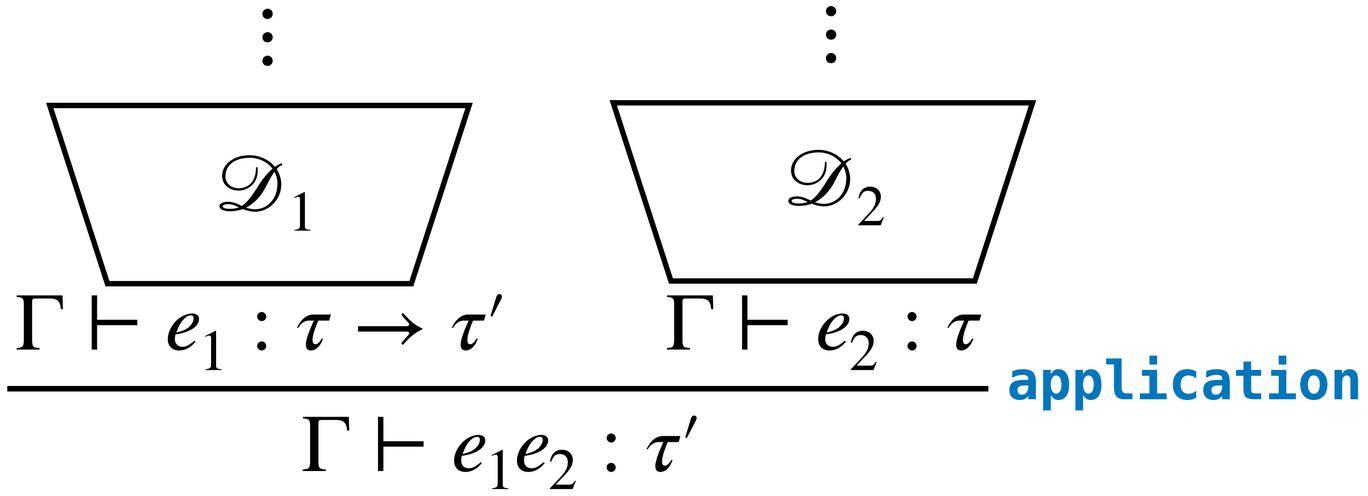
We need to show that expressions typed using just axioms satisfy well-scopedness

Inductive Hypothesis



If $e_1, \ldots, e_k$ are well-scoped (because they are typeable in the each of their contexts)

Inductive Step 1: Application



What if the last rule I applied was application?

Inductive Step 2: Abstraction

$$\dfrac{\begin{array}{c}\vdots \\ \mathscr{D} \\ \Gamma, x : \tau \vdash e : \tau'\end{array}}{\Gamma \vdash \lambda x^{\tau} . e : \tau \to \tau'}\ \textbf{abstraction}$$

What if the last rule I applied was abstraction?

Summary

Type systems delineate well-behaved expressions

We prove **meta-theoretic** properties about types systems to demonstrate this